Code No: 135CU **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, August - 2022
NON-CONVENTIONAL POWER GENERATION
(Common to CE, ME, ECE, CSE)

Time: 3 Hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1.a) With a neat sketch, explain the working principle of Pyranometer.
b) Define the following terms:
i) Altitude angle
ii) Incident angle
iii) Zenith angle. [8+7]

2.a) With the help of a neat sketch, explain the working of solar air heating system.
b) What are the advantages and disadvantages of concentrating collectors over flat plate collectors? [8+7]

3.a) Explain with diagram, how a Photovoltaic cell works.
b) What are the advantages and disadvantages of solar photovoltaic energy conversion? [8+7]

4.a) Define fill factor and explain the voltage-current characteristics of a solar cell.
b) Discuss the method of MPPT system. [8+7]

5.a) Describe the main consideration for selecting a site for wind generator.
b) Derive an expression for the total power of a wind stream. [8+7]

6.a) Compare horizontal and vertical axis windmill.
b) Explain with neat sketch, the working of a wind energy conversion system with main components. [7+8]

7.a) Explain the constructional detail and working of KVIC biogas plant.
b) Describe a hot dry rock geothermal resource power plant. [8+7]

8.a) With a neat sketch, explain the principle of open cycle OTEC system.
b) What are the advantages and disadvantages of a fuel cell? [8+7]

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